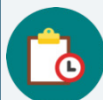


14.3 Two-Way Tables and Segmented Bar Graphs

Lesson Introduction

 Benchmarks	MA.912.DP.3.1 Construct a two-way frequency table summarizing bivariate categorical data. Interpret joint and marginal frequencies, and determine possible associations in terms of a real-world context.			 Learning Targets	- I can correctly match a segmented bar graph and data in a two-way table. - I can interpret a graphical display for bivariate categorical data.	
 Benchmark Coherence	Building On		Working Toward			
	MA.8.DP.1.1 Given a set of real-world, bivariate numerical data, construct a scatter plot or a line graph as appropriate for the context.		MA.912.DP.1.2 Within the Probability and Statics course, instruction includes the use of a spreadsheet and technology.			
 Essential Questions	- How can you identify which segmented bar graph matches the data in a two-way frequency table? - How can you interpret a graphical display for bivariate categorical data?					
 Lesson Narrative	This lesson continues to build students' understanding of two-way frequency tables from lessons 1 and 2. In this lesson, students will analyze how bivariate categorical data displayed in a two-way frequency table can also be displayed using a segmented bar graph. Students do this through an Exploration activity where students analyze an example before moving onto making matches of the same data, displayed in a two-way frequency table and segmented bar graph. Once students have a foundation of how the data is displayed, they move to interpreting data from a segmented bar graph.					
 Component Summary	14.3.1 Warm-Up (5 min.)	Notice and Wonder about segmented bar graphs (<i>SE p. 316</i>)		14.3.3 Bring It Together (15 min.)	Represent bivariate categorical data with segmented bar graphs (<i>SE p. 319</i>)	
	14.3.2 Exploration (15 min.)	Explore segmented bar graphs (<i>SE p. 317</i>)		14.3.4 Try It (10 min.)	Interpret segmented bar graphs (<i>SE p. 320</i>)	
	14.3.5 Wrap-Up (5 min.)	Ticket Out the Door on two-way frequency tables and segmented bar graphs (<i>SE p. 320</i>)				
 Resources	Glossary		Materials		Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> Segmented Bar Graph		(Optional) Copies of the pages with Tables 1-3 and Graphs A-D from 14.3.2 Exploration		(Optional) Turn question 1 in 14.3.2 Exploration into a card sort.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not realize that the bar totals in the 14.3.2 Exploration can display the marginal frequencies for the row or column. - Students may think their corrected statement 14.3.4 Try It is not correct if it does not match what their partner or teacher has.	- Consider having a completed, matching, two-way frequency table and segmented bar graph matches for 14.3.2 Exploration so that deeper conversations about how the data is displayed can occur. - When correcting the false statement in 14.3.4 Try It, consider having students share any corrected statements that were not shared, so that students can get feedback on whether their interpretation for the corrected statement is true.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share For 14.3.4 Try It, consider using a Think-Pair-Share routine to engage students in a conversation with their peers about why a statement is true or false. It will also allow them to have a conversation about how to correctly interpret the data displayed in a segmented bar graph.	MA.K12.MTR.2.1: Multiple Representations This MTR states to help students make connections between concepts and representations. During 14.3.2 Exploration and 14.3.3 Bring It Together, students take their understanding on how to construct and interpret two-way frequency tables, and apply that to how the data is represented in a segmented bar graph.
 Differentiate Instruction	Consider turning 14.3.2 Exploration into a card sort activity to allow students to physically hold and move the two-way frequency tables and the corresponding segmented bar graphs. By doing this, it will allow students to see the table and graphs side-by-side, so that they can easier see how the same data is displayed in both.	
Lesson Notes:		

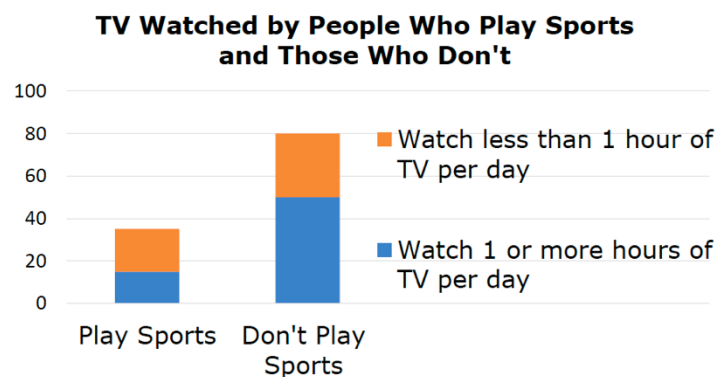
14.3.1 Warm-Up: Notice and Wonder

Component Overview: Students will answer two questions about what they notice and wonder about the segmented bar graph.



14.3.1 Warm-Up: Notice and Wonder

A segmented bar graph is shown.



1. What do you notice about the display?

Answers will vary.

2. What do you wonder?

Answers will vary.



During this lesson, students will be exploring segmented bar graphs, so refrain from answering any questions during 14.3.1 Warm-Up. Consider having students return to question 2 at the end of the lesson to see if what they wondered had been answered during the lesson.

14.3.2 Exploration: Segmented Bar Graphs

Component Overview: During 14.3.2 Exploration, students will first analyze how bivariate categorical data can be displayed in a two-way frequency table or a segmented bar graph, then work with a partner to match two-way frequency tables to the segmented bar graph that represents the same data.



14.3.2 Exploration: Segmented Bar Graphs

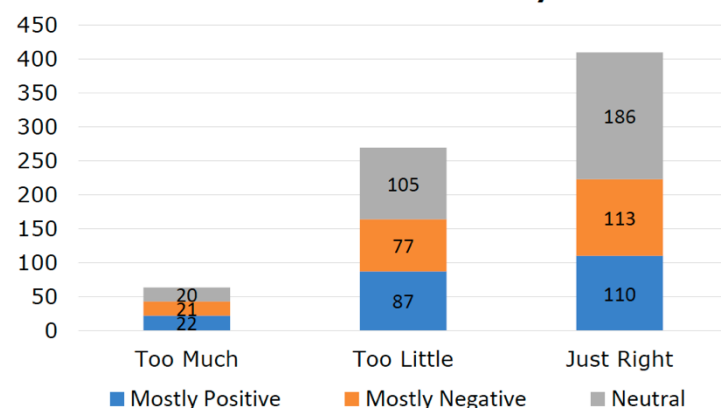
Bivariate categorical data can be displayed in a two-way frequency table as shown:

Teen Social Media Survey

		Effect on People Your Age			
Time spent With Friends in Person		Mostly Positive	Mostly Negative	Neutral	Total
	Too Much	22	21	20	63
	Too Little	87	77	105	269
	Just Right	110	113	186	409
	Total	219	211	311	741

It can also be displayed in a **segmented bar graph**.

Teen Social Media Survey



Consider providing an opportunity for students to analyze how the data in the two-way frequency table is displayed in the segmented bar graph by allowing a few minutes for them to individually study each display.



To encourage mathematical reasoning, consider providing the following questions for students to think about as they analyze the display:

- What do you notice about how the values are displayed in both?
- What do you notice about the labels in both displays?

Refrain from giving answers, because students will be exploring how data is shown in both during this activity. These are just suggestions for students to consider while analyzing both displays.

14.3.2 Exploration: Segmented Bar Graphs (continued)

Three two-way frequency tables and four segmented bar graphs are shown.

Table 1
Peoples' Social Media Preference by Age Group

	Less Than 18 Years Old	18 – 30 Years Old	Over 30 Years Old	Total
Snapchat	147	72	90	309
Facebook	23	56	112	191
Total	170	128	202	500

Table 2
Points Scored During a Volleyball Match

	University of Miami, Florida	Jacksonville University, Florida	Total
Spike Kills	18	22	40
Service Aces	39	48	87
Other	20	24	44
Total	77	94	171

Table 3
Emotional States Related to Meditation Practice

	Meditated	Did not Meditate	Total
Calm	52	6	58
Excited	11	24	35
Agitated	9	15	24
Total	72	45	117

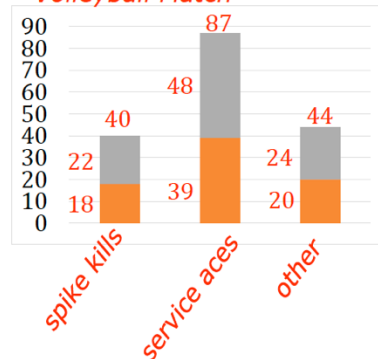


Consider making copies of the information on this page and having it available for students outside of their book so that they can refer to it for the activity, without having to flip back in the book repeatedly to review the data or grab information that they may need.

14.3.2 Exploration: Segmented Bar Graphs (continued)

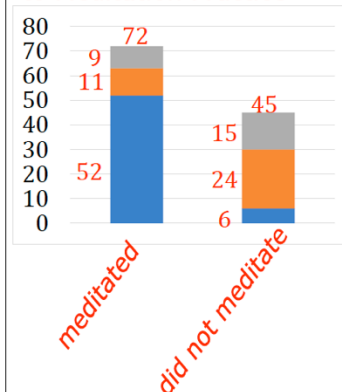
Graph A

Points Scored During a Volleyball Match



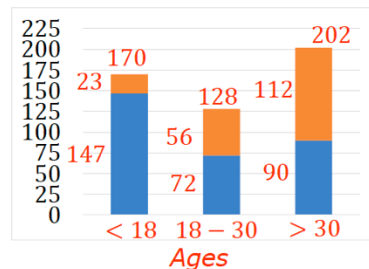
Graph B

Emotional States Related to Meditation Practice

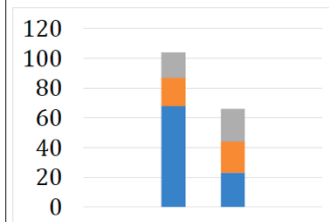


Students may notice that the segmented bar graphs are not labeled. They will be labeling these based on their matches with the two-way frequency tables during question 2.

Graph C *People's Social Media Preferences by Age*



Graph D



Consider making copies of the information on this page and having it available for students outside of their book, so that they can refer to it throughout 14.3.2 Exploration without having to flip back in the book repeatedly to review the data or grab information that they may need.

14.3.2 Exploration: Segmented Bar Graphs (continued)

1. Work with your partner to match each two-way frequency table to the segmented bar graph that displays the same data.

Two-Way Frequency Table	Segmented Bar Graph
1	☐ A ☐ B ☒ C ☐ D
2	☒ A ☐ B ☐ C ☐ D
3	☐ A ☒ B ☐ C ☐ D

2. Work with your partner to label the bars and their sections in the segmented bar graphs so that the categories represented by each bar are indicated.
The labeling is on the graphs above.
3. Discuss with your partner how joint frequencies are displayed in a segmented bar graph. Summarize your discussion.
Answers will vary. Joint frequencies are the pieces of the bar that show how the people in each group are categorized.



To allow for students to see the two-way frequency tables and segment bar graphs side-by-side, consider turning question 1 into a card sort by having pre-cut, two-way frequency tables and segmented bar graphs from the previous two pages.



Allow students to move onto question 2 without correcting any incorrect matches, as students might discover the error during question 2.



Allow for students to have this discussion in question 3 with their partner, then move onto 14.3.3 Bring It Together, as it will allow students to share what they have summarized in 14.3.3 Bring It Together, question 3.

14.3.3 Bring It Together: Representing Bivariate Categorical Data With Segmented Bar Graphs

Component Overview: During this activity, students are provided the opportunity to share what they learned during 14.3.2 Exploration.



14.3.3 Bring it Together: Representing Bivariate Categorical Data With Segmented Bar Graphs

Another way to represent the bivariate categorical data in a two-way frequency table is with a segmented bar graph.

1. In a two-way frequency table, marginal frequencies are

found in the

☐ center
☒ margins

 of the table. These numbers

express the

☒ totals
☐ parts

 of each category.

2. Carolyn concluded that marginal frequencies are represented by the height of each bar on a segmented bar graph.

Bailey argued that the marginal frequencies are represented by the sum of the segment heights with the same color for each color.

Which student do you agree with? Explain your reasoning.

Answers will vary. I agree with both students because the bar heights are the marginal frequencies for one set of variables and the sum of the segments of each color are the marginal frequencies for the other set of variables.

3. In a two-way frequency table, joint frequencies are

found in the

☒ center
☐ margins

 of the table. Joint

frequencies are represented by the

- ☐ total height of a bar.
 - ☒ segments of different colors within each bar.



As students provide responses to questions 1-3, consider having a display of students' work from 14.3.2 Exploration, or completed matches displayed so that students can justify their reasoning. Consider further probing questions into how the marginal frequencies are displayed in a segmented bar graph:

- How are the marginal frequencies displayed differently on the segmented bar graph for Tables 2 and 3 from 14.3.2 Exploration?
- How are the joint frequencies displayed differently on the segmented bar graph for Tables 2 and 3 from 14.3.2 Exploration?



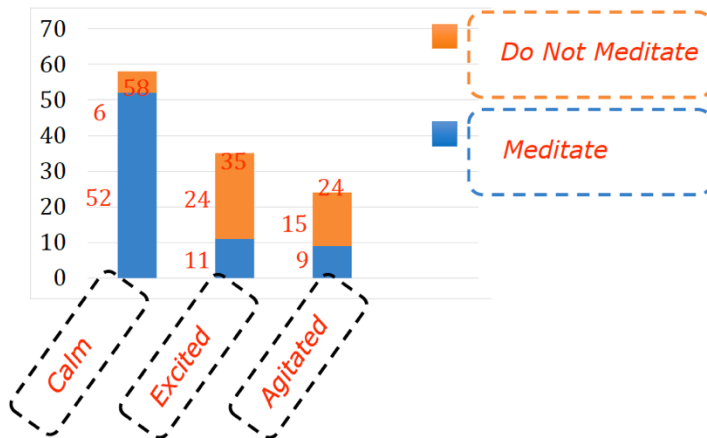
Consider allowing students to check question 3 with what they summarized with their partner in question 3 of the 14.3.2 Exploration.

14.3.3 Bring It Together: Representing Bivariate Categorical Data With Segmented Bar Graphs (continued)

4. Consider the data in the table and segmented bar graph.

People's Emotional States Related to Meditation Practice

	Meditated	Do not Meditate	Total
Calm	52	6	58
Excited	11	24	35
Agitated	9	15	24
Total	72	45	117



- Circle the marginal frequencies in the table.
- Label each marginal frequency in the dotted boxes.
- Label the value for each joint frequency.



Position students as sense makers of question 4. Ask students to take 2 minutes to read and think about their responses for questions 4a – c before asking for their responses. Consider the following questions to have students justify their thinking:

- How did you determine where to label each marginal frequencies in the graph?
- How did you determine where to label each value for each joint frequency?

14.3.4 Try It: Interpreting Segmented Bar Graphs

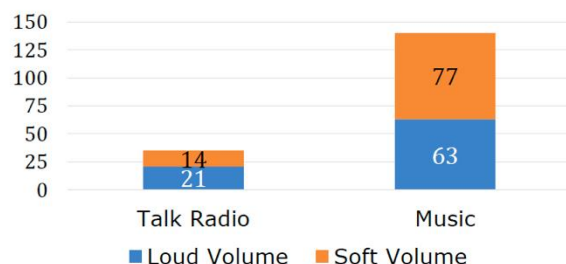
Component Overview: Students will be provided the opportunity to interpret joint and marginal frequencies using the segmented bar graph to determine if the provided statements are true or false.



14.3.4 Try It: Interpreting Segmented Bar Graphs

- Long-haul truckers were recently surveyed at a rest area in Tallahassee, Florida. The truckers were asked about what and how they primarily listen to audio in their trucks while on the road.

Long-Haul Truck Driver Listening Preferences



Determine whether each of the following statements is true or false. If the statement is false, correct the statement to make it true.

14 of those surveyed prefer loud volume. <i>84 of those surveyed prefer loud volume.</i>	☐ True ☐ False
140 of those surveyed prefer soft volume. <i>91 of those surveyed prefer soft volume.</i>	☐ True ☐ False
63 of those surveyed prefer music at a soft volume. <i>63 of those surveyed prefer music at a loud volume.</i>	☐ True ☐ False
More people prefer music than talk radio.	☐ True ☐ False
More people prefer talk radio to be at a loud volume than at a soft volume.	☐ True ☐ False



Think-Pair-Share

Consider implementing a Think-Pair-Share routine here. Allow students 5 minutes to solve each true or false statement individually. Then pair students to share whether they selected true or false as well as their reasoning. Lastly, have students share their solutions and reasoning as a whole class.



Answers could vary when correcting the false statements. Ensure that the corrected statements written by students are a correct interpretation.

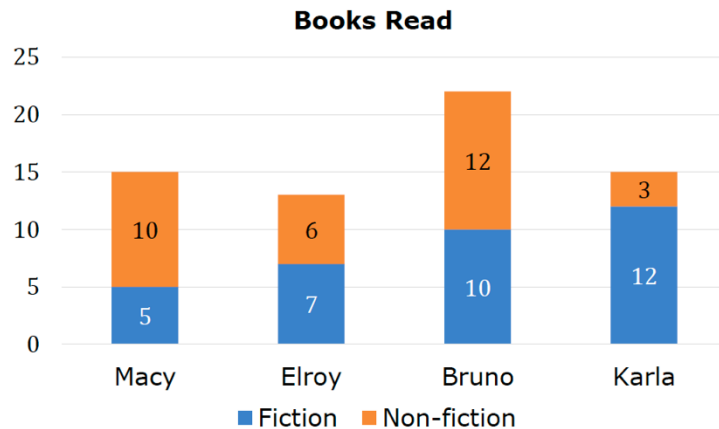
14.3.5 Wrap-Up: Ticket Out the Door

Component Overview: Students complete a Ticket Out the Door as a formative assessment to determine level of student understanding for the learning target, “I can interpret a graphical display for bivariate categorical data.”



14.3.5 Wrap-Up: Ticket Out the Door

Macy, Elroy, Bruno, and Karla each kept track of the books they read for a year. Their data is represented in the segmented bar graph shown. Use the segmented bar graph to answer the questions.



- What does the value 7 mean in this context?
The 7 on Elroy's bar stands for 7 fiction books. Elroy read 7 fiction books throughout the year.
- How many fiction books were read?
34 fiction books were read throughout the year.



Consider using data from this formative assessment to determine how to form small groups in Lesson 4.



Consider our online resources for additional remediation opportunities for students to practice or test their abilities.